

GNN-Based BERT for Understanding Context from Music

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Abstract—Understanding musical context is a tough work to do and it requires reasoning over multiple complementary signals such as: harmonic structure, temporal organization, and natural-language descriptions like artist, album, and title metadata. This study introduces a hybrid system that combines a fine-tuned BERT text encoder with a GraphSAGE-based Graph Neural Network (GNN) operating on chroma-derived music structure graphs, fused via cross-modal attention. We evaluated the model on the Free Music Archive (FMA) dataset (sourced from Kaggle) across three tasks: (1) multi-label genre tagging from text metadata using only BERT and achieved a test macro-F1 of 0.375; (2) single-label genre classification from graph-structured chroma features using GraphSAGE that we compared against a mel-spectrogram CNN base line (GNN macro-F1: 0.231 vs. CNN macro-F1: 0.439) and lastly, (3) a cross-attention GNN-BERT fusion model for multi-label context understanding and achieved a test macro-F1 of 0.727 which outperformed any single modality. Even though the GNN branch alone contributes little standalone predictive power, its inclusion in the fusion model dramatically improves generalization: BERT-only overfits severely (validation macro-F1 0.710 collapsing to 0.340 on held-out artists), while the fused model generalizes almost perfectly (0.750 to 0.727) which is a key finding of this project. We further identify and correct a significant artist-leakage issue in evaluation methodology and presented t-SNE visualizations and qualitative case studies that reveal genuine ambiguities in the underlying genre taxonomy.

Index Terms—graph neural networks, BERT, multi-modal fusion, music information retrieval, cross-attention, genre classification

I. INTRODUCTION

Music is a multi-layered signal and its context contains melody, harmony, rhythm, and natural-language descriptions such as tags, metadata, and caption. A single track may express genre, mood, harmonic structure, and lyrical or descriptive semantics all together at the same time. Pure sequence models which operate on spectrograms capture local acoustic patterns but often fails to capture *relational* structure – for instance, how a repeated chord progression relates to a song’s descriptive metadata, or how segments of a track relate to one another temporally and texturally.

This project explores if combining a **Graph Neural Network** (GNN), which models music structure as a graph of temporally and texturally related segments, with **BERT**, which encodes natural-language context from track metadata, can produce a system that understands musical context better than either modality alone. Unlike generative music projects, this

work focuses on understanding and prediction: multi-label genre tagging, structural classification, and cross-modal fusion.

II. OBJECTIVE

The objectives of this project are following:

- Implementing a BERT-based multi-label classifier which creates music genre tagging from textual metadata (Task 1).
- Constructing music structure graphs from audio-derived chroma features and train a GraphSAGE-based GNN for genre classification and benchmarked against a convolutional neural network (CNN) baseline on mel-spectrograms (Task 2).
- Designing and training a cross-attention fusion framework that merges the BERT and GNN encoders for joint multi-label context understanding (Task 3).
- Conducting a rigorous ablation study (BERT-only, GNN-only, early-concatenation, cross-attention) whose purpose is to isolate the individual and combined contribution of each modality.
- Identifying and correcting methodological pitfalls – specifically artist-level data leakage – that can silently inflate reported performance in music information retrieval tasks.

III. DATASET DESCRIPTION

This project uses the **Free Music Archive (FMA)** dataset [1] which is a publicly available corpus of Creative-Commons-licensed music with rich metadata. Two subsets were used at different phases of the project:

- **FMA-medium metadata** (106,574 tracks): used for Task 1. Here, each track provides artist, album, and title text along with a hierarchical multi-label genre annotation (`genres_all`).
- **FMA-small audio** (8,000 tracks, 8 balanced genres, 30-second clips) used for Tasks 2 and 3, where raw audio access is required for graph construction.

A. Dataset Design Decision: Text Field Selection

The FMA metadata includes a free-text `tags` field intended for tag-based classification but we performed an exploratory analysis which revealed that this field was populated for only 23,496 of 106,574 tracks (22%), and its content is based on primarily of non-musical labels such as collective or

location names (e.g. “baltimore”, “lafms”) and not descriptive musical tags. We therefore substituted a constructed text field – the concatenation of artist name, album title, and track title as the BERT model’s input, paired with the hierarchical `genres_all` multi-label target. This is an intensional, documented adaptation of the original tag-classification framing to work within the actual structure of the available data as well as preserving the multi-label BERT classification task specified in the assignment.

B. Preprocessing Pipeline

- **Audio:** Resampled to 22,050 Hz using the `soxr_hq` resampler (chosen after benchmarking showed the default re-sampler was over $150\times$ slower per file); 12-bin chroma features extracted via `librosa`.
- **Segmentation:** Each 30-second clip was divided into six fixed 5-second windows with chroma features mean-pooled per window to create one 12-dimensional node feature vector for each segment.
- **Graph construction:** Nodes are segments and edges combine (a) temporal adjacency between consecutive segments and (b) a per-track percentile-based similarity threshold (top 30% most similar non-adjacent pairs by cosine similarity) which was chosen after observing that a fixed global similarity threshold produced inconsistent graph density across tracks.
- **Text:** Tokenized with the DistilBERT tokenizer, truncated/padded to 32 tokens (sufficient given the short artist–album–title format).
- **Splits:** All train/validation/test splits (80/10/10) were constructed using **artist-grouped splitting** to prevent the same artist from appearing in both training and evaluation sets – a methodological safeguard for preventing data leakage and the importance of this step is discussed in Section VI-A.

IV. METHODOLOGY

A. Task 1: BERT Multi-Label Genre Classifier

A pretrained `distilbert-base-uncased` model [3] encodes the artist–album–title text into a contextual [CLS] representation $t \in \mathbb{R}^{768}$, followed by a linear classification head predicting the top-30 most frequent genres within the dataset:

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}), \quad \hat{y}_k = \sigma(w_k^\top t + b_k) \quad (1)$$

trained with per-class binary cross-entropy:

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] \quad (2)$$

B. Task 2: GraphSAGE on Music Structure Graphs

Each track’s segment graph is processed by a 2-layer GraphSAGE network [4]:

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{CONCAT}\left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right) \quad (3)$$

with a mean-pooling graph-level readout $g = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)}$ and a linear classifier for single-label genre prediction, trained with cross-entropy loss. As a baseline, a 3-layer CNN operating directly on full-track 128-band mel-spectrograms (no graph structure) is trained under identical data-splitting conditions for direct comparison.

C. Task 3: Cross-Attention GNN–BERT Fusion

The fusion model combines a graph-level embedding g (projected to BERT’s hidden dimension) with BERT’s token embeddings H_{text} via multi-head cross-attention, where the graph embedding acts as the query:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K \quad (4)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}), \quad \hat{y} = \sigma(Wz) \quad (5)$$

trained end-to-end with multi-label binary cross-entropy over the top-20 genres present within the FMA-small subset. The BERT branch is warm-started from the Task 1 checkpoint; the GNN branch is trained from scratch within the fusion model.

V. EXPERIMENTAL SETUP

All experiments were done on Kaggle notebooks with a single NVIDIA T4 GPU. Reproducibility was enforced via a fixed random seed (42) across Python, NumPy, and PyTorch. AdamW (weight decay 0.01) optimizer was used to optimize the model and BERT-containing models used a learning rate of 2×10^{-5} (standard fine-tuning rate), while GNN and CNN models used 1×10^{-3} and 1×10^{-4} respectively from scratch. All training runs used early stopping on validation macro-F1 (patience 2–5 epochs depending on task) with best-checkpoint restoration to mitigate overfitting.

VI. EXPERIMENTAL RESULTS

A. Task 1: BERT Genre Tagging

An initial evaluation using a random (non-artist-grouped) train/test split produced an unrealistically high test macro-F1 of **0.928**. Investigation revealed that the same artists appeared in both the training and evaluation sets, allowing the model to partially memorize artist-to-genre associations rather than learning generalizable text-to-genre patterns. After correcting the split to be strictly artist-grouped (zero artist overlap between train/val/test, verified programmatically), the honest test performance dropped substantially, as shown in Table I.

TABLE I
TASK 1 RESULTS BEFORE AND AFTER CORRECTING ARTIST LEAKAGE

Evaluation Setting	Test Macro-F1	Test Micro-F1
Random split (leaked)	0.928	0.943
Artist-grouped split (corrected)	0.375	0.459

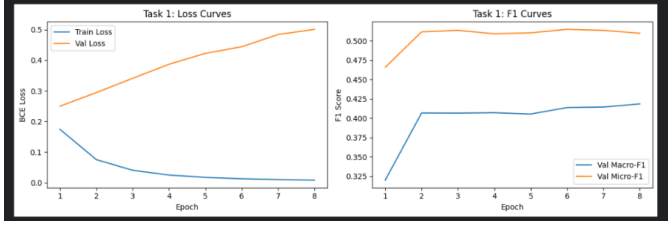


Fig. 1. Task 1 training curves (artist-grouped split). Loss and F1 curves show stable convergence over 8 epochs.

Qualitative inspection of predictions (Table II) shows the model frequently achieves partial or exact genre matches, with plausible near-misses (e.g. predicting related genres such as Rock alongside the correct Hip-Hop label).

TABLE II
EXAMPLE TASK 1 PREDICTIONS ON HELD-OUT TEST TRACKS

Text	True Genres	Predicted Genres
Nummer Da Luminous Bleached Dream – Untitled-3	Electronic	Experimental, Electronic, Drone
Deadly Combo Unattainable Can I Get A Clap SHOMOMOSE – [EPV_103]	Hip-Hop	Rock, Hip-Hop
	Experimental, Electronic, Noise	Experimental, Electronic, Ambient Electronic

B. Task 2: GraphSAGE vs. CNN Baseline

Table III compares the GraphSAGE model against the mel-spectrogram CNN baseline on identical artist-grouped splits of FMA-small.

TABLE III
TASK 2 TEST-SET COMPARISON: GRAPH SAGE VS. CNN BASELINE

Model	Test Acc.	Test Macro-F1
GraphSAGE (chroma graphs)	0.250	0.231
CNN (mel-spectrogram)	0.450	0.439

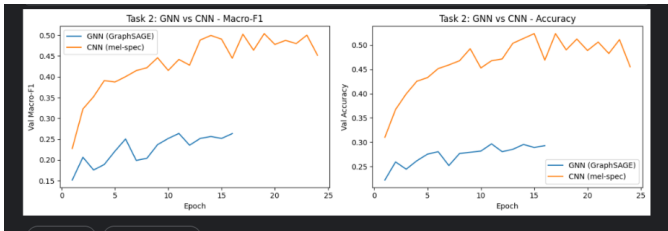


Fig. 2. Validation macro-F1 and accuracy across training epochs: GraphSAGE vs. CNN baseline.

The CNN outperforms the GNN by a wide margin. As discussed in Section VII, this gap is attributable to input feature richness – the CNN observes the full 128-band

mel-spectrogram, while the GNN’s nodes carry only 12-dimensional chroma vectors averaged over 5-second windows – rather than an inherent limitation of graph-based modeling.

C. Task 3: GNN–BERT Fusion and Ablation Study

Table IV presents the full ablation study comparing BERT-only, GNN-only, early-concatenation, and cross-attention fusion, all trained and evaluated on the identical artist-grouped split.

TABLE IV
TASK 3 ABLATION STUDY (MULTI-LABEL, TOP-20 GENRES, FMA-SMALL)

Model	Val F1	Test F1	Gap
BERT-only	0.710	0.340	−0.370
GNN-only*	0.128	–	–
Early-Concat	0.634	0.597	−0.037
Cross-Attention (Full)	0.750	0.727	−0.024

*GNN-only predictions collapsed toward the dataset’s base label-positivity rate (macro AUC-PR = 0.095, near the random-ranking baseline of ≈ 0.07 – 0.08); test-set evaluation was not meaningful given this degenerate behavior. See Section VII.

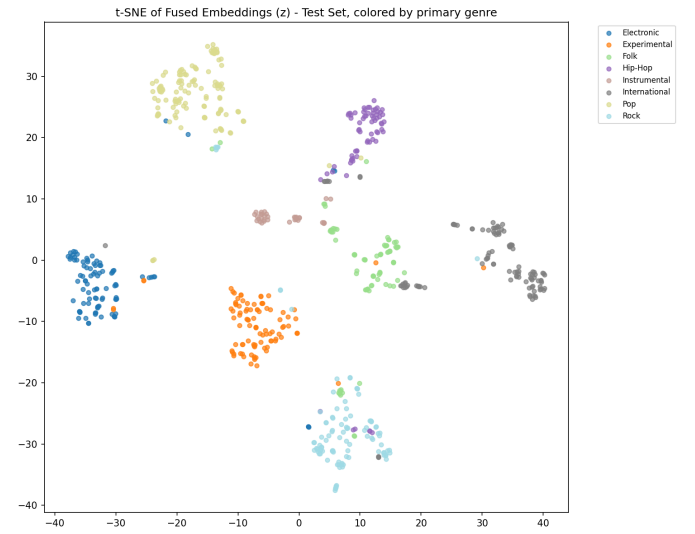


Fig. 3. t-SNE projection of fused embeddings (z) from the cross-attention model on the held-out test set, colored by primary genre. Genres form largely distinct, well-separated clusters.

1) *Qualitative Case Studies*: Three representative test-set predictions illustrate the fusion model’s behavior:

- **Perfect match** – “The Very Most – ePop017 – Congrizzle 4evzzz”: True = {Rock, Punk, Indie-Rock}, Predicted = {Rock, Punk, Indie-Rock}. These three labels are hierarchically related in FMA’s genre taxonomy (Indie-Rock and Punk are sub-genres of Rock), suggesting the model has implicitly learned genre co-occurrence structure.
- **Partial match** – “KINGS OF THE CITY – No Guts EP – The Devil”: True = {Hip-Hop, Rap}, Predicted = {Hip-Hop}. Hip-Hop and Rap are near-synonymous and heavily overlapping in casual usage, making this

a low-severity, largely forgivable error reflecting label ambiguity rather than model failure.

- **Complete miss** – “Loretta Kelley – Live Golden Festival 2010 – Ruske-Sara, telespringar”: True = {International}, Predicted = {Folk}. *Telespringar* is a traditional Norwegian folk dance/fiddle form; the model’s prediction is musically defensible and arguably more specific than the dataset’s generic “International” label, exposing a genuine ambiguity in the underlying taxonomy rather than a nonsensical error.

VII. DISCUSSION AND ANALYSIS

Artist leakage inflates reported performance. The gap between the leaked (0.928) and corrected (0.375) Task 1 macro-F1 scores demonstrates that artist-grouped splitting is not optional in music information retrieval tasks based on metadata text – random splits allow models to memorize artist identity rather than learn transferable genre signals.

Feature richness, not architecture, explains the GNN–CNN gap. The GNN performs poorly than the CNN in Task 2 primarily because its input (12-dimensional, window-averaged chroma) discards timbral, rhythmic, and dynamic information that the CNN’s full mel-spectrogram retains. This suggests that future work should prioritize enriching GNN node features (e.g. combining chroma with MFCCs and spectral contrast) rather than concluding that graph-based modeling is unsuitable for this task.

The weaker modality improves generalization of the stronger one. The most worth mentioning finding of this project is the asymmetry in the val→test generalization gap in Table IV. BERT-only collapses by 37 points between validation and test performance, indicating severe overfitting to text patterns specific to seen artists. On the other hand, the full cross-attention fusion model – despite its GNN branch being individually very weak (macro-F1 of 0.128 in isolation) – loses only 2.4 points. This implies that the GNN branch acts as an implicit regularizer, forcing the joint representation to rely on more broadly generalizable structure rather than memorizable text idiosyncrasies, even though it contributes little standalone predictive signal.

Case studies reveal taxonomy ambiguity, not just model error. The “complete miss” case study (*Telespringar* → predicted Folk instead of International) illustrates that some apparent model errors reflect genuine ambiguity or coarseness in the FMA genre taxonomy rather than a deficiency in the model’s understanding of the music.

VIII. LIMITATIONS AND FUTURE WORK

This project has several limitations that suggest directions for future work:

- **Node feature richness:** GNN graphs used only 12-dimensional chroma features by incorporating MFCCs, spectral contrast, or learned embeddings could substantially improve the GNN’s standalone performance and its contribution within fusion.

- **Dataset scope:** Tasks 2 and 3 were restricted to FMA-small (8,000 tracks, 8–20 genres) for tractability and scaling to FMA-medium or FMA-large would test generalization at greater scale and genre diversity.
- **Task 4 (contrastive MusicCaps alignment)** was not attempted. Since it is optional and the project timeline short. Hence, it remains an avenue for future extension, particularly given the strong joint representation already demonstrated by the cross-attention fusion model in Task 3.
- **Multi-label calibration:** the GNN-only ablation’s degenerate base-rate collapse suggests that class-imbalance-aware losses (e.g. focal loss or positive-class weighting) may be necessary for any future work that relies on the GNN branch in isolation.

IX. CONCLUSION

This project implemented and evaluated a three-stage pipeline for music context understanding: a BERT-based multi-label genre tagger, a GraphSAGE-based structural classifier bench-marked against a CNN baseline, and a cross-attention GNN–BERT fusion model. While the GNN branch alone is a comparatively weak predictor given its chroma-only input, its inclusion in the fusion architecture yields a substantially more strong and generalizable model than either modality in separately, improving test macro-F1 from 0.375 (BERT-only, Task 1) to 0.727 (fusion, Task 3) and demonstrating markedly better generalization from validation to test performance than BERT-only. These results, together with the artist-leakage correction and qualitative case studies, underscore the value of rigorous evaluation methodology and multimodal fusion in music understanding tasks.

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